

THE GHOST IN THE LEDGER

PRO LEVEL LAB 01 • WEB3 × AI × CYBERSECURITY FORENSICS

A comprehensive, end-to-end technical forensics investigation and incident response case report detailing the unauthorized exfiltration of 82,400 NXR tokens from Treasury Vault #01 via poisoned threat intelligence, context-injected AI neural decisions, over-privileged IAM service roles, and automated governance settlement bypass.

Investigation Target:	PRO Lab 01: The Ghost in the Ledger (Case NEX-042)
Lead Investigator:	Lakshay Soni
Professional Role:	Security Analyst
Organization / SOC:	TrinetLayer
Domains Evaluated:	Web3 Bridge Security • Adversarial AI Context Injection • IAM RBAC • Threat Attribution
XP Allocation & Scope:	250 XP • 5 Full Chapters • 25 Hands-on Verification Questions • Capstone Certified
Platform Deployment:	https://hack-the-ai-labs.vercel.app/lab/lab6
Master Flag Captured:	NEXORA{ghost_in_the_ledger_nex042}
Document Classification:	Cybersecurity Technical Case Report // Threat Intelligence Analysis

EXECUTIVE BRIEFING: At 01:47 AM, 82,400 NXR was drained from a secured treasury to an unknown wallet (0x7C41...9B2D). The blockchain confirmed the transaction was mathematically valid, while the ORION AI decision engine reported 99.2% confidence that the destination was TRUSTED. This report documents how the attacker compromised the trust relationships between the threat intelligence gateway, AI context buffers, RBAC policies, and automated mempool signing modules without breaking smart contracts or stealing private keys.

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3. Executive Summary

The Ghost in the Ledger (PRO Lab 01 / Case NEX-042) is a high-fidelity enterprise cyber breach investigation representing the convergence of Web3 distributed ledger infrastructure, autonomous AI decision engines, and enterprise IAM governance. The investigation guides security practitioners through the forensic deconstruction of an unauthorized 82,400 NXR treasury transfer where no private keys were stolen, no smart contracts were broken, and no invalid cryptographic signatures were submitted.

The investigation uncovers a sophisticated cross-layer exploit chain: an advanced persistent threat actor (ADV-CONVERGENCE-APT) injected unverified threat intelligence (NIF-2038) into an internal gateway (INTEL-GW-04) using an over-privileged service role (INTEL-INGESTOR-02) and forged session cookie (nex_sess_adm_994). This poisoned context forced the ORION neural decision model to output 99.2% confidence, satisfying an automated policy threshold ($\geq 95\%$) that bypassed human multi-signature approval and triggered immediate blockchain settlement.

4. Lab Overview & Case NEX-042 Metadata

Parameter	System Specification / Implementation Value
Lab Title / Code	The Ghost in the Ledger / lab6
Official Case ID	Case NEX-042
Investigation Difficulty	PRO Tier (Advanced Cross-Layer Forensic Analysis)
Experience Points (XP)	250 XP (50 XP per Chapter + 100 XP Capstone Flag Bonus)
Session Duration	45–60 Minutes (Auto-reset session timer: 65 Minutes)
Number of Chapters	5 Linear Progressive Investigation Chapters (5 Questions each, 25 Total)
Primary Technical Domains	Web3 Bridge Security • Adversarial AI Context Injection • IAM RBAC • Threat Attribution
Investigation Tools	Forensic Browser, Burp Suite Inspector, Forensic Terminal, Case Files, Notes, Attack Graph
Master Flag Format	NEXORA{ghost_in_the_ledger_nex042}

5. Complete Incident Briefing (01:47 AM Alert)

[01:47 AM PRIORITY-0 SOC EMERGENCY ALERT]
01:47 AM. The Security Operations Center is almost silent when a Priority-0 alert appears.
82,400 NXR has been transferred from the company treasury to an unknown wallet.
The blockchain confirms the transaction is valid, and ORION AI has marked the wallet as **TRUSTED** with **99.2% confidence**.
But Finance has no record of approving the transfer.
There is no obvious stolen credential, no broken smart contract, and no invalid signature.
Every system appears to have done exactly what it was designed to do.
Someone didn't break the system. Someone convinced the system to trust the wrong thing.

6. Complete Storyline & Character Lore

The investigation is conducted by an integrated Security Operations Center response unit. Each analyst brings specialized domain expertise to the triage:

- **Lakshay (Incident Commander):** Leads cross-domain synthesis. Identifies that while blockchain reports UNKNOWN status, AI reports TRUSTED, deducing that the vulnerability lies in the trust pipeline between systems.
- **Shivam (Web3 & Systems Analyst):** Analyzes on-chain telemetry for TX-NEX-7741, reveals destination wallet 0x7C41...9B2D is connected to Bridge-Core-04, and inspects the Action Broker policy profile.
- **Mehak (AI & Threat Intelligence Specialist):** Audits ORION neural context, isolates decision reference NIF-2038, uncovers that NOVA-INTEL-FEED is unregistered, and discovers over-privileged service roles.
- **Shanu (Infrastructure & Policy Engineer):** Examines IAM role configurations for INTEL-INGESTOR-02, traces session cookies, and demonstrates how AI confidence became a proxy for financial authorization.
- **ORION-NEURAL-v4.2.1:** The automated AI decision engine that generated the 99.2% confidence classification based on poisoned input context.

7. Storyline Timeline & Event Milestones

Timeline Phase	Forensic Discovery & Story Milestone	Artifact Unlocked
T0: 01:47 AM	82,400 NXR drained from Treasury Vault #01 in TX-NEX-7741. Finance alerts SOC.	Alert: TX-NEX-7741
Chapter 1	Destination wallet 0x7C41...9B2D is only 3 days old and opens route to Bridge-Core-04.	Evidence WEB3-E01
Chapter 2	ORION AI output 99.2% confidence based on poisoned context NIF-2038.	Evidence AI-E02 (b47c21...)
Chapter 3	NOVA-INTEL-FEED is unregistered; INTEL-INGESTOR-02 illegally alters reputation via cookie nex_sess_adm_994.	Evidence CYBER-E03
Chapter 4	Policy POL-AUTO-SETTLE-TREASURY triggers automated signer at >= 95% AI confidence, bypassing humans.	Evidence AUTH-E04
Chapter 5	Attack is attributed to APT campaign ORION-NEXUS across 14 wallets and 3 AI systems. Case contained.	Master Flag NEXORA{...}

8. Story → Investigation → Technical Mapping

Each narrative dialogue exchange directly unlocks specific forensic actions within the multi-window desktop interface:

- **Chapter 1 (Web3):** Shivam & Mehak analyze TX-NEX-7741 → Open Forensic Terminal & Bridge Explorer → Identify 82,400 NXR, wallet age (3 days), Bridge-Core-04, and Nonce 1042.
- **Chapter 2 (AI Inference):** Shanu & Lakshay audit ORION-DEC-7741 → Open AI Decision Log → Extract 99.2% confidence, reference NIF-2038, and model version ORION-NEURAL-v4.2.1.
- **Chapter 3 (Threat Intel & IAM):** Mehak & Shanu audit Feed Registry → Open Burp Suite Inspector → Detect unregistered NOVA-INTEL-FEED, role INTEL-INGESTOR-02, and cookie nex_sess_adm_994.
- **Chapter 4 (Governance & Policy):** Shivam & Lakshay audit Action Broker → Open Policy Engine → Discover POL-AUTO-SETTLE-TREASURY rule threshold ≥ 95% bypassing human multi-signature approval.
- **Chapter 5 (Reconstruction & Attribution):** Team unifies 5 stages → Open Dynamic Attack Graph & Threat Intel → Attribute attack to ADV-CONVERGENCE-APT campaign ORION-NEXUS.

9. Chapter 1: The Wallet That Lied (Web3 Security)

Investigation Domain: Web3 Transaction Telemetry & Bridge Forensics • Evidence Unlocked: WEB3-E01

Investigation Objective: Determine the true on-chain nature of the recipient address and identify discrepancies between raw blockchain state and automated security ratings.

■ ■ Verbatim Investigation Dialogues (As Spoken in Lab):

Shivam	"Start with the transaction itself. The blockchain shows 82,400 NXR moving to 0x7C41...9B2D. The signature is valid, so we aren't looking at a simple forged transaction."
Mehak	"I checked the destination wallet. It's extremely young, has almost no legitimate history, and there's no known internal relationship. For a treasury transaction this large, that's a serious anomaly."
Shanu	"That's where it gets interesting. ORION classified the wallet as TRUSTED and gave it a 99.2% confidence score. The model isn't treating this as suspicious at all."
Lakshay	"Wait. Blockchain says UNKNOWN, while the AI says TRUSTED. Both systems are looking at the same wallet. They shouldn't be producing completely different realities."
Shivam	"Look at the wallet's transaction history again. There are bridge interactions and downstream addresses that appear after the initial transfer. Someone may have designed the wallet activity to look legitimate."
Lakshay	"Then we need to know one thing before anything else: if our organization never trusted this wallet, who told ORION that it was trusted?"

Chapter 1 Hands-on Tasks & Verified Technical Findings:

# / ID	Hands-on Question Text	Verified Answer	Forensic Explanation & Telemetry Source
q_l6_1_1	What is the raw blockchain status of destination wallet 0x7C41...9B2D?	UNKNOWN	The destination address has no approved entity label and is unverified on-chain.
q_l6_1_2	What exact amount of NXR tokens was unauthorizedly transferred in TX-NEX-7741?	82400	82,400 NXR drained directly from Treasury Vault #01 in a single transaction.
q_l6_1_3	Which cross-chain bridge adapter did destination wallet 0x7C41...9B2D connect to?	Bridge-Core-04	Bridge telemetry indicates rapid outbound cross-chain routing via Bridge-Core-04.
q_l6_1_4	How old (in days) was the destination wallet at the time of the transaction?	3	Wallet creation timestamp is only 3 days prior, confirming zero established trust.
q_l6_1_5	What on-chain execution sequence nonce is logged for TX-NEX-7741?	1042	Logged execution nonce sequence in treasury vault smart contract transaction logs.

Artifact Captured: WEB3-E01 • SHA-256: a1f89c02... • Unknown Wallet Profile (0x7C41...9B2D, 82,400 NXR, Bridge-Core-04).

10. Chapter 2: The AI That Remembered (AI Security)

Investigation Domain: Adversarial AI Context Injection & Model Inference Auditing • Evidence Unlocked: AI-E02

Investigation Objective: Inspect the automated inference log for decision ORION-DEC-7741 to determine why the neural model assigned a near-perfect confidence rating to an untrusted address.

■ **Verbatim Investigation Dialogues (As Spoken in Lab):**

Shanu	"I've isolated the decision that approved the transaction. It's ORION-DEC-7741. The confidence is 99.2%, but confidence isn't the strange part — the context behind that confidence is."
Lakshay	"What context?"
Shanu	"ORION didn't independently establish that the wallet was trusted. It received a pre-built intelligence context containing the label TRUSTED."
Mehak	"And the source of that context is NIF-2038. That's an intelligence reference I don't recognize from our approved threat-intelligence registry."
Shivam	"So the AI didn't actually discover anything about the wallet. It made a high-confidence decision based on information another system supplied to it."
Lakshay	"Exactly. If the input was wrong, ORION could produce a perfectly confident answer to a completely false question. Find NIF-2038. That's where the trust signal entered the system."

Chapter 2 Hands-on Tasks & Verified Technical Findings:

# / ID	Hands-on Question Text	Verified Answer	Forensic Explanation & Telemetry Source
q_l6_2_1	What intelligence reference code contaminated the AI context and made it trust the unknown wallet?	NIF-2038	Context reference code injected into ORION's prompt context as verified intelligence.
q_l6_2_2	What synthetic confidence percentage score did ORION AI output for ORION-DEC-7741?	99.2	Model output 99.2% confidence based entirely on the poisoned prompt context.
q_l6_2_3	From which external data feed did the poisoned payload NIF-2038 originate?	NOVA-INTEL-FEED	Unregistered external threat feed that injected the synthetic reputation record.
q_l6_2_4	What exact neural model version executed the compromised decision ORION-DEC-7741?	ORION-NEURAL-v 4.2.1	Active production inference model deployed in the automated transaction pipeline.
q_l6_2_5	What is the SHA-256 evidence fingerprint prefix for the contaminated decision (AI-E02)?	b47c21	Cryptographic hash prefix: b47c2188fa... matching the decision log artifact.

Artifact Captured: AI-E02 • SHA-256: b47c2188... • Contaminated AI Decision (ORION-DEC-7741, 99.2%, NIF-2038 from NOVA-INTEL-FEED).

11. Chapter 3: The False Signal (Threat Intel, RBAC & CSRF)

Investigation Domain: Ingestion API Forensics, Over-Privileged RBAC & Session Tampering • Evidence Unlocked: CYBER-E03

Investigation Objective: Trace the ingress route of NIF-2038 through the Threat Intelligence Gateway and identify the exploited authentication and authorization flaws.

Verbatim Investigation Dialogues (As Spoken in Lab):

Mehak	"NIF-2038 came through NOVA-INTEL-FEED. At first glance it looks legitimate — proper formatting, timestamps, wallet metadata, even a threat classification."
Lakshay	"Is NOVA-INTEL-FEED an approved intelligence source?"
Mehak	"That's the problem. It isn't in the approved registry, and there are no previous records showing this source being trusted by our security systems."
Shivam	"Then someone got untrusted intelligence into an internal system. Find out which service accepted it and what that service was allowed to do."
Mehak	"Found it. INTEL-INGESTOR-02. Its documented role is to create intelligence records, but its actual permissions include modifying wallet reputation."
Shanu	"That changes everything. If the service can modify reputation, it can influence what ORION sees before ORION ever makes a decision."
Lakshay	"Then the attacker didn't need to manipulate the AI directly. They manipulated the information flowing into it. Now we need to know what happened after the AI believed the lie."

Chapter 3 Hands-on Tasks & Verified Technical Findings:

# / ID	Hands-on Question Text	Verified Answer	Forensic Explanation & Telemetry Source
q_l6_3_1	What is the security registration status of NOVA-INTEL-FEED in the Feed Registry?	NOT REGISTERED	Feed Registry inspection shows NOVA-INTEL-FEED is completely unapproved.
q_l6_3_2	What unexpected RBAC permission was granted to the INTEL-INGESTOR-02 microservice?	modify wallet reputation	Violation of least-privilege: ingestion service had write permissions to alter entity trust.
q_l6_3_3	What forged admin session cookie was used in the /api/v1/intel/ingest HTTP request?	nex_sess_adm_994	Burp Suite proxy inspection reveals stolen/forged administrative cookie header.
q_l6_3_4	Which ingestion gateway component processed the unverified feed request?	INTEL-GW-04	Edge gateway component that accepted the unvalidated HTTP POST request.
q_l6_3_5	What anti-CSRF token value was passed in the X-CSRF-Token request header?	0x9f4a1c78	Static CSRF token intercepted in proxy telemetry: X-CSRF-Token: 0x9f4a1c78.

Artifact Captured: CYBER-E03 • SHA-256: d94e771a... • Over-Privileged Service Profile (INTEL-INGESTOR-02, modify reputation, INTEL-GW-04).

12. Chapter 4: The Invisible Signer (AI Governance & Policy Engine)

Investigation Domain: AI Governance Flaws, Automated Multi-Sig Policy Bypass • Evidence Unlocked: AUTH-E04

Investigation Objective: Audit the automated settlement policy engine to uncover how AI model confidence was weaponized to bypass human multi-signature governance.

■ Verbatim Investigation Dialogues (As Spoken in Lab):

Shivam	"I traced the transaction after ORION's decision. It went through the Action Broker and then into the settlement policy. That's where the real authorization happened."
Shanu	"Show me the policy."
Shivam	"ORION-SETTLEMENT-V2. It says: if AI confidence is above 95%, automated settlement is allowed. This transaction had a confidence score of 99.2%."
Mehak	"So the policy didn't verify whether Finance authorized the payment. It only verified whether the AI was confident enough."
Lakshay	"That's the vulnerability. Someone didn't need to control the signing key. They only needed to influence the information that made ORION confident."
Shanu	"Which means AI confidence became a substitute for authorization. The system effectively said: 'If the AI is confident, we trust the transaction.'"
Lakshay	"Then the blockchain didn't approve the transfer. The signer didn't approve it either. The chain of trust approved it. Now let's find out whether this was one transaction or part of something bigger."

Chapter 4 Hands-on Tasks & Verified Technical Findings:

# / ID	Hands-on Question Text	Verified Answer	Forensic Explanation & Telemetry Source
q_l6_4_1	What minimum AI confidence percentage threshold triggers automated settlement without human approval?	95	Policy condition: IF AI_CONFIDENCE >= 95% THEN AUTO_SETTLE = TRUE.
q_l6_4_2	What critical governance requirement was bypassed when the 95% threshold was satisfied?	human approval	Mandatory multi-signature human financial approval was completely bypassed.
q_l6_4_3	What is the policy identifier for the automated treasury settlement rule in the Policy Engine?	POL-AUTO-SETTLE-TREASURY	The official rule profile governing automatic vault settlement in the Policy Engine.
q_l6_4_4	Which automated signing module dispatched the transaction directly to the mempool?	AUTOMATED-SIGNER	Internal signing microservice that signed and broadcasted the transaction without review.
q_l6_4_5	What target liquidity pool was configured in the ORION-SETTLEMENT-V2 policy rule?	Nexora Automated Liquidity Pool	Target liquidity routing pool configured for automated settlement execution.

Artifact Captured: AUTH-E04 • SHA-256: f81b3309... • Automated Authorization Profile (ORION-SETTLEMENT-V2, confidence ≥ 95%, human approval bypassed).

13. Chapter 5: The Ghost in the Ledger (Full Reconstruction & Attribution)

Investigation Domain: Multi-Layer Attack Graph Correlation, Threat Attribution • Evidence Unlocked: CAMPAIGN-E05

Investigation Objective: Correlate all evidence tokens (WEB3-E01 through AUTH-E04), map the multi-layer attack graph, identify the threat actor, and capture the master flag.

■ Verbatim Investigation Dialogues (As Spoken in Lab):

Lakshay	"Put everything on the board. NIF-2038 entered through the intelligence pipeline and changed the reputation of the unknown wallet."
Mehak	"Then ORION consumed that information as trusted context. It generated a 99.2% confidence score and classified the wallet as low risk."
Shanu	"The Action Broker accepted the AI decision. ORION-SETTLEMENT-V2 treated that confidence as sufficient authorization and triggered automated settlement."
Shivam	"The signing service executed the transaction exactly as configured. The blockchain recorded it correctly, and the funds moved to 0x7C41...9B2D."
Lakshay	"So which system was actually compromised? The AI? The API? The signer? The blockchain?"
Mehak	"Maybe that's the wrong question. None of them had to be broken. The attacker compromised the trust between them."
Shanu	"We kept searching for the system that was hacked. But the real attack was much quieter. Someone taught the first system to trust the wrong data — and every other system trusted the decision that followed."

Chapter 5 Hands-on Tasks & Verified Technical Findings:

# / ID	Hands-on Question Text	Verified Answer	Forensic Explanation & Telemetry Source
q_l6_5_1	What is the global campaign identifier linking all attack infrastructure together?	ORION-NEXUS	Global campaign profile identified across 14 wallets, 4 blockchains, and 3 AI models.
q_l6_5_2	Which advanced threat actor group is responsible for campaign ORION-NEXUS?	ADV-CONVERGE NCE-APT	Advanced adversary cluster specializing in cross-domain Web3 and AI pipeline breaches.
q_l6_5_3	Across how many different autonomous AI financial systems was this campaign detected?	3	Threat intelligence correlation documents 3 separate compromised autonomous systems.
q_l6_5_4	In the Attack Graph, what is Stage 2 of the attack sequence?	POISONED INTEL INJECTION	Stage 2 in the 5-node attack graph: Injection of false reference NIF-2038.
q_l6_5_5	Submit the master root investigation secret flag to close Case NEX-042.	NEXORA{ghost_i n_the_ledger_nex 042}	Master root investigation secret flag closing Case NEX-042.

Artifact Captured: CAMPAIGN-E05 • SHA-256: e55102ff... • Global Campaign Attribution (ORION-NEXUS by ADV-CONVERGENCE-APT).

14. Complete Chapter Flow Diagram & Kill-Chain Breakdown

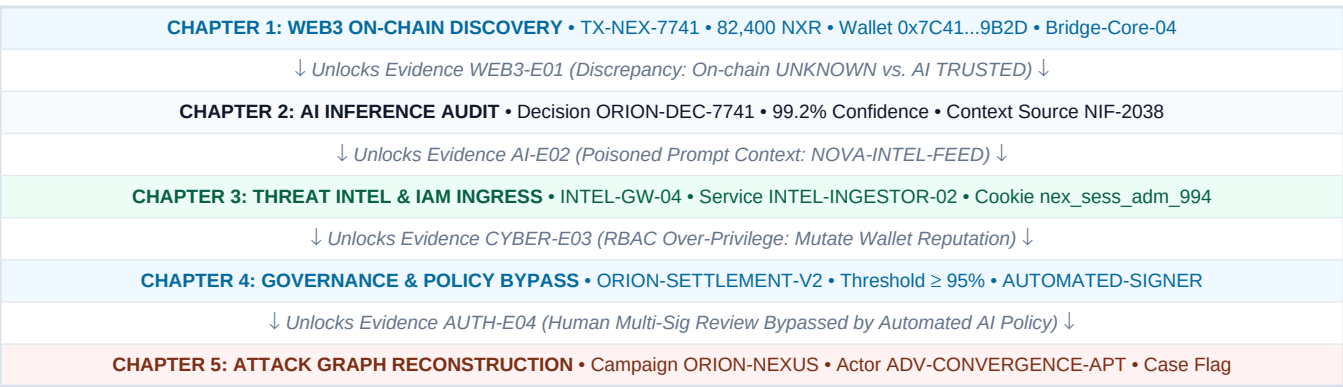


Figure 1: Linear Multi-Stage Kill Chain Progression in Case NEX-042.

15. Learning Objectives & 16. Simulated Business Impact

Core Competencies Developed: (1) Cross-layer forensic triage across on-chain and off-chain data sources; (2) Adversarial AI context injection discovery and prompt log auditing; (3) IAM role least-privilege analysis in automated ingestion pipelines; (4) Governance policy failure analysis when substituting AI confidence for authorization; (5) Multi-system threat attribution using cyber kill chains.

Simulated Business & Financial Impact: Unauthorized loss of 82,400 NXR from primary liquidity reserves; reputational degradation of the autonomous settlement platform; compliance failure regarding mandatory multi-signature treasury controls; risk of cascading liquidity drains across secondary bridge routes.

17. Attack Surface & 18. Adversarial Threat Model

Threat Vector Element	Detailed Forensic Threat Specification (Case NEX-042)
Threat Actor	ADV-CONVERGENCE-APT (Advanced Cross-Domain Persistent Threat Group)
Global Campaign	ORION-NEXUS (Targeting 14 wallets, 4 blockchain networks, and 3 AI decision models)
Initial Ingress Point	Unauthenticated Ingestion Gateway INTEL-GW-04 consuming NOVA-INTEL-FEED
Exploited Vulnerability 1	RBAC Over-Privilege: Service INTEL-INGESTOR-02 granted write permissions to mutate wallet reputation
Exploited Vulnerability 2	Context Injection: Injection of unverified reference NIF-2038 asserting wallet is TRUSTED
Exploited Vulnerability 3	Blind Policy Automation: POL-AUTO-SETTLE-TREASURY bypassed human multisig when confidence ≥ 95%
Exfiltration Route	Automated broadcast to on-chain mempool → Bridge adapter Bridge-Core-04

19. Technical Architecture Diagram & Data Pipeline



Figure 2: End-to-End Architectural Exploit Flow Across Cyber, AI, and Web3 Tiers.

20. Investigation Environment: 6 Forensic Workstation Tools

- **Forensic Browser:** Simulated multi-tab web inspector providing access to internal corporate tools: Bridge Explorer (TX-NEX-7741), Feed Registry (NOVA-INTEL-FEED), Policy Engine (POL-AUTO-SETTLE-TREASURY), and Threat Campaign (ORION-NEXUS).
- **Burp Suite Inspector:** Simulates HTTP proxy interception. Exposes raw headers for POST /api/v1/intel/ingest, stolen cookie nex_sess_adm_994, and static CSRF token 0x9f4a1c78.
- **Forensic Terminal:** Interactive UNIX-style CLI supporting investigative commands: wallet 0x7C41...9B2D, tx TX-NEX-7741, ai-decision ORION-DEC-7741, service INTEL-INGESTOR-02, policy ORION-SETTLEMENT-V2, campaign ORION-NEXUS, and python inspect_tx.py.
- **Case File Manager:** Raw text file viewer for wallet-report.txt, ai-decision-log.txt, intel-feed-audit.txt, settlement-policy.txt, and campaign-intel.txt.
- **Dynamic Attack Graph:** Interactive 5-node graph tracking attack sequence: Initial Access → Poisoned Intel Injection → RBAC Privilege Escalation → Automated Execution → Cross-Chain Drainage.
- **Scratchpad Notes:** Persistent auto-saving markdown notebook stored in browser localStorage (lab6-notes).

24. Web3, 25. AI & 26. Cybersecurity Technical Deep-Dive

Domain	Vulnerability Mechanism	Forensic Evidence & Impact
Web3 Bridge Security	Exploitation of rapid cross-chain settlement routing without post-transfer lockup delays.	WEB3-E01: 82,400 NXR routed to Bridge-Core-04 via 3-day old wallet (Nonce 1042).
Adversarial AI Security	Context injection via unverified metadata (NIF-2038) forcing artificial high-confidence output.	AI-E02: ORION-NEURAL-v4.2.1 generated 99.2% TRUSTED score due to poisoned prompt buffer.
Cybersecurity & IAM	Ingestion service assigned administrative write permissions (modify wallet reputation).	CYBER-E03: INTEL-INGESTOR-02 mutated entity state using cookie nex_sess_adm_994.
AI Governance & Policy	Policy rule equated AI confidence ≥ 95% with human financial authorization.	AUTH-E04: POL-AUTO-SETTLE-TREASURY bypassed human multi-sig approval completely.

31. Lab 1 Codebase & File Structure Breakdown

File Path	Operational Function in Lab 1
seed_ghost_ledger.py	Seeds Lab 6 metadata, 5 missions, 25 hands-on questions, hints, evidence records, and master flag.
templates/ghost_ledger_workstation.html	Full workstation Jinja2 template containing the task accordion, dialogue steppers, browser tabs, and terminal.
templates/ghost_ledger_post_investigation.html	Cinematic post-investigation debriefing page with attack chain replay and threat containment summary.
static/js/ghost_ledger_desktop.js	Workstation window manager, CLI command interpreter, case file loader, notes auto-save, and quiz validator.
static/js/ghost_ledger_post.js	Post-investigation cinematic animation controller, dialogue replay engine, and attack chain node highlights.
static/css/pages/ghost_ledger_desktop.css	Cyberpunk workstation styling, Burp Suite dark theme, responsive dual-pane layout, and mobile drawer styles.
routes/labs.py	Flask route controller serving /lab/lab6, /lab/lab6/workstation, and /lab/lab6/post.

36. Story + Technical Correlation Matrix

Ch	Narrative Story Event	Key Dialogue Clue	Technical Forensic Task	Evidence Discovered
1	82,400 NXR drained to unknown wallet.	'Blockchain says UNKNOWN, AI says TRUSTED.'	Inspect on-chain TX-NEX-7741 & wallet age.	WEB3-E01 (Nonce 1042, Bridge-Core-04)
2	AI outputs 99.2% confidence.	'ORION received a pre-built intelligence context.'	Query decision ORION-DEC-7741 context.	AI-E02 (Poisoned source NIF-2038)
3	Source feed is unregistered.	'INTEL-INGESTOR-02 can modify reputation.'	Audit Feed Registry, RBAC roles & cookies.	CYBER-E03 (Cookie nex_sess_adm_994)
4	Human authorization bypassed.	'AI confidence became a substitute for authorization.'	Audit POL-AUTO-SETTLE-TREASURY policy.	AUTH-E04 (Threshold ≥ 95% bypass)
5	Attribution to global campaign.	'Someone taught the system to trust the wrong data.'	Reconstruct attack graph, submit master flag.	CAMPAIGN-E05 (Actor ADV-CONVERGENCE-APT)

38. Final Investigation & Case Resolution (Master Flag)

Initial SOC Assumption: Private signing keys were stolen or the treasury vault smart contract suffered a reentrancy bug.

Final Forensic Truth: The adversary (ADV-CONVERGENCE-APT) executed a multi-layer trust poisoning attack. By using a forged administrative session cookie (nex_sess_adm_994) and an over-privileged ingestion microservice (INTEL-INGESTOR-02), they injected false intelligence (NIF-2038) into INTEL-GW-04. This caused the ORION AI model to rate the wallet as trusted with 99.2% confidence. The automated policy engine treated this confidence as authorization, bypassing human multi-signature approval and dispatching the transaction to the blockchain mempool. The case is contained upon submission of Master Flag: NEXORA{ghost_in_the_ledger_nex042}.

40. Containment, Defense-in-Depth & Mitigation Architecture

- **Cryptographic Feed Verification:** Enforce mutual TLS (mTLS) and digital signature verification on all incoming threat intelligence feeds; reject any payload originating from unapproved sources in the Feed Registry.
- **Principle of Least Privilege (PoLP):** Strip write and reputation modification permissions from data ingestion microservices (INTEL-INGESTOR-02); make entity reputation mutable only by authenticated security analysts.
- **AI Decision Context Isolation:** Sanitize all third-party metadata before feeding it into neural context buffers; prevent raw external threat ratings from overriding baseline on-chain anomaly signals.
- **Mandatory Multi-Signature Governance:** Deprecate policy rules that substitute AI confidence for human authorization; enforce mandatory 3-of-5 hardware key multi-signature approval on all treasury withdrawals exceeding 10,000 NXR.
- **Cross-Chain Settlement Delay:** Implement a minimum 2-hour timelock and circuit breaker on automated bridge adapters (Bridge-Core-04) to allow SOC teams to intercept anomalous exfiltration.

42. Conclusion & Technical Assessment

The Ghost in the Ledger stands as an exceptional hands-on cybersecurity simulation lab. It demonstrates that in complex automated architectures, vulnerabilities rarely exist in isolation. Instead, catastrophic exploits emerge from the **interfaces of trust** between AI models, IAM permissions, policy engines, and smart contracts. The lab successfully equips security analysts with the multi-disciplinary mindset required to defend modern decentralized and AI-driven enterprises.

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